

4. CRYOSTAT

Preparation of frozen tissue sections at low temperature for rapid histological evaluation.

SOP No.: LAB-HISTO-04	Revision: 01	Effective: _____
Department: Histopathology / Research	Prepared by: _____	Approved by: _____

Purpose

- To cut thin sections from rapidly frozen tissue when rapid intraoperative assessment or preservation of specific tissue components is required.

PPE & Safety

- Wear lab coat, gloves and eye/face protection as required. Use approved cryostat practices for potentially infectious fresh tissue. Beware of very cold surfaces and sharp blades. Never reach into the chamber near an exposed blade. Follow the laboratory's decontamination protocol.

Materials

- Fresh/frozen tissue; specimen chuck; embedding medium suitable for cryostat use; cryostat blade; labelled slides; forceps/brush; approved disinfectant.

Procedure

- Verify chamber temperature and instrument condition. Set the validated temperature appropriate to tissue type and embedding medium.
- Place tissue in embedding medium on the chuck and freeze rapidly using the instrument's approved method.
- Secure the chuck and blade. Lock the handwheel during setup and whenever accessing the blade area.
- Trim the block, then cut sections at the validated thickness. Transfer suitable sections directly to labelled slides.
- Proceed immediately to the laboratory's approved staining/processing workflow.
- After work, remove tissue debris and perform the validated cryostat decontamination cycle. Record use and cleaning.

Quality Control

- Sections should be intact and diagnostically representative, with minimal freezing artefact, tearing or curling. Confirm specimen identity at every transfer.

Troubleshooting

- Sections crumble → tissue/block may be too cold or improperly embedded. Sections curl → adjust temperature/anti-roll device and blade condition. Thick/thin variation → check blade, chuck stability and sectioning mechanism.

Critical Note

- Do not use a generic temperature or disinfectant schedule in place of the manufacturer's and laboratory's validated protocol.

Important: This is a general SOP template. Before operational use, insert the exact manufacturer model, validated temperatures/times, reagent names, decontamination procedure, QC acceptance criteria and institutional biosafety/chemical-safety requirements.